

Fig. 2: Power supply circuit

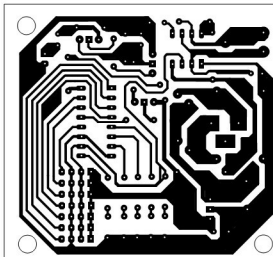


Fig. 3: A single-side, actual-size PCB layout for the circuit within the dotted lines for Christmas decoration light

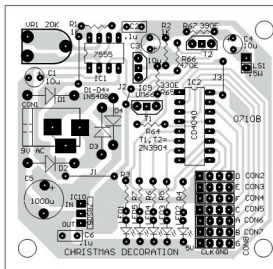


Fig. 4: Component layout for the PCB in Fig. 3

The first extension adds eight LEDs to the display. The LEDs light up one at a time until all eight are lit. These stay lit for a while, and then

turn off one at a time. This action repeats.

The second extension adds musical sound. It employs a special music generating IC, intended for use in musical greeting cards.

The music can be chosen to suit the occasion. For example, you can choose from "Jingle Bell," "Santa Claus is coming to town," "We wish you a Mary Christmas" and so on. The musical tone plays every few minutes.

The circuit allows flexibility in timing of the change in lighting, number of LEDs and their different colours.

Circuit description

Fig. 1 shows the circuit for LED-based Christmas decoration light. It comprises timer 7555, 14-stage ripple-carry binary counter CD4040, shift registers 74HC164, simple melody generator UM66, regulator 78S05, LEDs and a few discrete components. 7555 is configured as an astable multivibrator. Oscillation frequency is determined by resistor R1, preset VR1 and capacitor C1. Preset VR1 is used to set the frequency to about 1 Hz. Pin 5 of 7555 is bypassed to ground via capacitor C2 to prevent any noise from altering the calculated frequency.

Oscillation frequency is available at pin 3 of 7555. Pin 3 drives LED1, clock pin 10 of ripple counter CD4040 and clock pin 8 of shift register 74HC164. Ripple counter

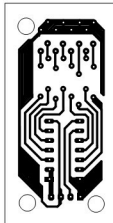


Fig. 5: A single-side, actual-size PCB layout for the circuit around single 74HC164 (IC3)

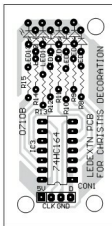


Fig. 6: Component layout for the PCB in Fig. 5

CD4040 drives LED2 through LED5 and remaining output pins are used for shift register 74HC164 and melody generator UM66. Remaining LEDs of the circuit are connected to the outputs of shift registers 74HC164. Output from pin 3 of IC2 is used as serial input for shift register IC3.

Shift register 74HC164 consists of a number of single-bit D-Type flip-flops connected together in a chain arrangement, so the output from one flip-flop becomes the input for the next flip-flop and so on, thereby

moving the stored data serially.

Generally, shift registers are used to store and convert data from either a serial to parallel format or parallel to serial format with all the internal flip-flops being driven by a common clock (CLK) signal making them synchronous devices. The first register of IC3 acquires the data from pin 3 of counter CD4040. Each time clock pin 8 goes high, the data in each register is shifted to the next register (available at the output of IC3). This shifting of register data is shown by LED6 through LED13. Shift registers IC4 through IC8 work similarly.

Output pin 13 of counter IC2 goes high and transistor T1 conducts to